



- Reduces to known problem in P
 - List the problem we solved?
 - string matching, shortest path, solving polynomials + finding coefficients

- Reduced to known problem in NP-Complete
 - Formulate as an MIP and try a MIP solver
 - Heuristics
 - Greedy
 - Approximation algorithms

- Solving unknown problem in P

- Examples:

http://uva.onlinejudge.org/index.php?option=com_onlinejudge&Itemid=8&category=9

3

- Data Structures needed?

List the data structures included with C#. Briefly describe each

Basic

- Int - 4 byte value that stores whole numbers
- double - 8 bytes, stores real numbers
- String - stores an array of characters
- byte - stores a value from 0 to 255, 1 byte value
- bool - stores either true or false
- long - 8 bytes, stores whole numbers
- float - 4 bytes, ranging from 1.5×10^{-45} to 3.4×10^{38} with a precision of 7 digits
- char - 2 byte value that represents a character in Unicode

Non Basic

- Stack: the latest data put on a stack is the first to come off
- List - objects linked together by one element pointing to the next
- Queue - the earliest data in the queue is the first to come out

- $\{ \dots \}$ - stores a list of objects
- HashMap - stores pairs of key-values. A key must be unique while the value does not.
- HashSet - stores similar objects as a set
- Array - stores a list of objects
- Dictionary - stores key-value pairs
- Vector - much like an array, but the length is not known
- $\{ \dots \}$ - stores a list of objects
- $\{ \dots \}$ - stores a list of objects

o Methodology

List the different techniques that can possibly be used to solve the problem.

- Complete Search: Brute Force, How big for $n!$, n^n , 2^n , $n C k$ or $n P k$
- Divide & Conquer - eliminating the search space of a problem to efficiently provide an answer, like binary search trees
- Dynamic Programming - breaking a big problem into smaller tasks
- Quantum Computing - provides classical answers much more efficiently than a classical algorithm can using quantum mechanics
- Greedy - makes the best choice with available data at that time
- Backtracking - eliminates possibilities from the final solution
- Hill climbing - starts with one solution & makes incremental changes until the optimal solution is found
- $\{ \dots \}$ - stores a list of objects

o Can you do better?

- Finding running time Big Oh
 - Recursive
 - o Masters Method
 - Non-Recursive
 - o Analyze the loops